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INFERENCES ON TRIP MATRICES FROM OBSERVATIONS ON LINK VOLUMES: A BAYESIAN STATISTICAL APPROACH

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Abstract—Previous methods for estimating a trip matrix from traffic volume counts have used the principles of maximum entropy and minimum information. These techniques implicitly give as little weight to prior information on the trip matrix as possible. The new method proposed here is based on Bayesian statistical inference and has several advantages over these earlier approaches. It allows complete flexibility in the degree of belief placed on the prior estimate of the trip matrix and also allows for different degrees of belief in different parts of the prior estimate. Furthermore under certain assumptions the method reduces to a simple updating scheme in which observations on the link flows successively modify the trip matrix. At the end of the scheme confidence intervals are available for the estimates of the trip matrix elements.

INTRODUCTION

The assignment process converts a knowledge of the trip matrix $[T_{ij}]$ into estimated link flows; the two main classes of assignment process are proportional assignment (in which the proportion of travellers between zones i and j who use link k, Q_{ij}^k, is independent of the T_{ij}) and equilibrium assignment (in which a link travel time is a function of the volume of traffic on that link and traffic is assigned in accordance with the equal travel time principle). The dual problem of determining a trip matrix from observations on link volumes has received much attention in the last few years, and is a problem of a different type; it is underspecified, in that p, the number of links on which measurements of the volume of traffic are made is generally very much less than n, the number of origin-destination pairs in the trip matrix. There are, therefore, multiple solutions to the problem. In order to determine a unique solution (the most likely solution, in some sense), some other information must be added: this is introduced in the form of *prior* information (for example, a target trip matrix $[T_{ij}^*]$). Then the combination of prior information and observations produces a unique *posterior* estimate of the trip matrix. Willumsen (1981) gives a useful general description of the problem.

Methods for producing trip matrices in this way have been proposed and described by Robillard (1975), Willumsen (1982), Van Zuylen and Willumsen (1980), Van Zuylen (1981), Van Vliet and Willumsen (1981), Hauer and Shin (1981) and Bell (1983), using the assumption of proportional assignment and by Nguyen (1977) and LeBlanc and Farhangian (1982) using equilibrium assignment. The problem of estimating turning flows at an intersection from counts on the inflows and outflows is, as will be shown later, of the same form and work by Jeffreys and Norman (1977), Mekky (1979), Van Zuylen (1979), Norman *et al.* (1979) and Cremer and Keller (1981) has been directed towards this problem.

In these previous studies a unique solution is ensured by defining some objective function, generally in the form of a "distance" between the trip matrix $[T_{ij}]$ and the prior or target trip matrix $[T_{ij}^*]$. Van Zuylen, for example, has used the principle of minimum information and Willumsen and Bell that of maximum entropy. As Van Zuylen and Willumsen point out, the general effect of these two principles is very similar, although the fitted models produced are slightly different in detail. The posterior trip matrix is an amalgam of the prior trip matrix and the observations, but in this amalgam as little weight as possible is given to the prior. Furthermore, in these previous methods only point values are specified in the prior; i.e. the matrix $[T_{ij}^*]$ is stated but no measure of the degree of belief in this prior is given or allowed for.

Now this problem of combining prior beliefs and observations to produce posterior beliefs is a standard one in Bayesian statistical inference (see, for example, Lindley and Smith, 1972). If one has 100% confidence in one's prior beliefs then no random observation, however remarkable, will change one's opinions and the posterior beliefs will be identical to the prior beliefs. If, on the other hand, one has little confidence in the prior, the observations will play the dominant role in determining the posterior beliefs. It is envisaged here that the prior information may come from an old transportation study or, in the case of the turning flows problem, short manual counts at the intersection. It is also envisaged that the observations made on the link volumes or approach road counts will be subject to random error and therefore neither of the two extreme cases described above will necessarily apply in practice. The uncertainties in the prior beliefs and the observations could be of comparable magnitude. An information minimising approach, then, could well be throwing away useful information in the prior.

Furthermore a greater degree of confidence may be held for some parts of the prior beliefs than for others. For example, a recent transportation study may have been carried out on part of the current study area and so some elements of the trip matrix may be known much more accurately than others; or, some turning flows at an intersection may have been counted accurately over a long time, or it may be that certain turning movements are banned or impossible. In these cases, then, there are varying degrees of belief in different parts of the prior.

The new method to be described here is based on Bayes' theorem, (see, for example, Lindley, 1971) which says that

$$p(\theta|y) = \frac{f(y|\theta)p(\theta)}{\int f(y|\theta)p(\theta)\,\mathrm{d}\theta} \tag{1}$$

in which $p(\theta)$ is the prior probability density of the parameter θ, $f(y|\theta)$ is the likelihood of the observation y and $p(\theta|y)$ is the posterior probability density of θ. The method, which is based on the assumption of proportional assignment, is easy to apply and allows for more flexibility than existing methods in the degree of belief in the prior and also in the possibility of different degrees of belief in different parts of the prior. It will be seen that, in the special case of minimal confidence in the prior, the results from this statistical method may be similar to those from the information minimising or entropy maximising approaches.

STATEMENT OF THE PROBLEM

Let the unknown flow from zone i to zone j in a transportation network be T_{ij}, and let the known proportion of this flow which uses link k be Q_{ij}^k. Then the total flow on link k will be $\sum_i \sum_j Q_{ij}^k T_{ij}$. Suppose that a count is made on the flow on link k and that this count is a_k. Assuming for the moment that this count is made without error (although this assumption will be relaxed later), this means that there is a constraint of the form

$$\sum_i \sum_j Q_{ij}^k T_{ij} = a_k \tag{2}$$

for each observed link flow. In practice the number of observed link flows, p, will be very much less than the number of origin–destination pairs, n, and so the problem of inferring the n T_{ij}'s from the p a_k's is underspecified: there are many solutions. The mathematical structure of this problem may be written in the following form: let there be n parameters $\theta_1, \ldots \theta_n$, and let there be observations $g_1, \ldots g_p$ made on the values of p independent linear combinations of these parameters:

$$\sum_{j=1}^{n} h_{ij}\theta_j = g_i \quad (i = 1, \ldots p). \tag{3}$$

If $p < n$ there is no unique solution for the θ_j. (It is, of course possible that, because of inconsistencies amongst the h_{ij} and g_i, there is no feasible solution. However this possibility will not be pursued any further here.)

Before moving on to generalise this problem by relaxing some of the assumptions above and then to analyse it, let us consider another similar problem: that of estimating the turning flows at an intersection from observations on the inflows and outflows on the approach roads to the intersection. Let the turning flow from approach i to approach j be denoted by T_{ij}, and let the total flow going into the intersection on approach i be a_i whilst that coming out of the intersection on approach i is b_i. Then there are constraints of the form

$$\sum_j T_{ij} = a_i \quad \text{and} \quad \sum_j T_{ji} = b_i \tag{4}$$

for each approach i. Assuming that U turns are impossible, at a typical four arm intersection there will be 12 turning flows to be estimated by means of observations on 7 independent in- or out-flows (the eighth one must be a linear combination of the other seven since $\sum_i a_i$ must equal $\sum_i b_i$). Again it is not possible to determine the T_{ij} uniquely without some additional information. It can be seen that this problem is a special case of the trip matrix problem, with all $T_{ii} = 0$ and the Q_{ij}^k taking the values 0 or 1 only.

We shall assume that the additional information required to give a unique solution is in the form of prior beliefs about the parameters $\theta_1, \ldots \theta_n$ (the T_{ij}'s). Unlike previous methods, though, these prior beliefs will not be expressed in a single value for each parameter but in the form of a prior *distribution* on each parameter, with a specified mean and variance. Therefore if a parameter is known precisely its prior variance will be zero, whilst if it is wished to assume very little about its value a very large variance would be specified. Furthermore it will be assumed that there is, in general, random error in the observations $g_1, \ldots g_p$, and so there will be a dispersion (or covariance) matrix associated with these observations. This generalises the statement of the problem above. Of course if an observation on a link flow is made over such a long time as to ensure that the observation on the flow rate is effectively without error, then the variance of that flow will be zero. There will be, then, two sources of random variability: in the prior distributions of the θ_j and in the observations g_i on linear functions of the θ_j. The relative magnitude of these variabilities will determine whether the posterior distribution is more heavily influenced by the prior distribution or the observations.

THE METHOD OF BAYESIAN INFERENCE

Let $\boldsymbol{\theta}$ denote the column vector of parameters $(\theta_1, \ldots \theta_n)$ of length n, $\mathbf{g}$ be the column vector $(g_1, \ldots g_p)$ of observations and $\mathbf{H}$ be the $p \times n$ matrix whose element h_{ij} is (in the trip matrix problem) the proportion of flow between O-D pair j which uses link i ($i = 1, \ldots p$; $j = 1, \ldots n$). It is assumed that $\mathbf{H}$ is known without error. The observation $\mathbf{g}$ is made on the random vector $\mathbf{Y}$ which is $\mathbf{H}\boldsymbol{\theta} + \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon}$ is a vector of random measurement errors with zero mean and therefore Bayes' theorem says that since the denominator in (1) does not include $\boldsymbol{\theta}$ the posterior density of $\boldsymbol{\theta}$ is proportional to the product of the likelihood of the observations from that $\boldsymbol{\theta}$ and the prior density of $\boldsymbol{\theta}$:

$$p(\boldsymbol{\theta}|\mathbf{g}) \propto f(\mathbf{g}|\boldsymbol{\theta})p(\boldsymbol{\theta}). \tag{5}$$

Although Bayes' theorem holds for any form of $p(\boldsymbol{\theta})$ and $f(\mathbf{g}|\boldsymbol{\theta})$, the tractability of the mathematics here depends on the distributional assumptions made for the priors and for the observations. Because the random variables are concerned with counts, the true distribution is probably some multivariate form of the Poisson distribution. For counts with means which are not too small, the multivariate normal (MVN) provides an accurate approximation; the approximation is not so good for small means. Since the method of proportional assignment is being assumed the mean value of $\mathbf{Y}$, $\mathbf{H}\boldsymbol{\theta}$, is a linear function of $\boldsymbol{\theta}$ and, if $\boldsymbol{\theta}$ is

MVN, $\mathbf{H}\boldsymbol{\theta}$ is MVN also. Despite some misgivings then about its use for cells with small means the multivariate normal seems to be the appropriate choice for the distribution of both $\boldsymbol{\theta}$ and the random errors $\boldsymbol{\epsilon}$, especially as it follows from these assumptions that the posterior distribution of $\boldsymbol{\theta}$ is also MVN. This will be shown in the following theorem.

Theorem. If $\boldsymbol{\theta} \sim MVN\ (\boldsymbol{\mu}_0, \mathbf{V}_0)$ *and* $\mathbf{Y} = \mathbf{H}\boldsymbol{\theta} + \boldsymbol{\epsilon}$ *where* $\boldsymbol{\epsilon} \sim MVN\ (\mathbf{0}, \boldsymbol{\Sigma})$, *so that* $\mathbf{Y} \sim MVN\ (\mathbf{H}\boldsymbol{\theta}, \boldsymbol{\Sigma})$, *it follows by Bayes' theorem than the posterior distribution of* $\boldsymbol{\theta}$ *given observations* $\mathbf{Y} = \mathbf{g}$ *is also MVN.*

The dispersion matrices $\mathbf{V}_0$ and $\boldsymbol{\Sigma}$ (whose main diagonal elements are variances, and whose other elements are the covariances of the variables) are of size $n \times n$ and $p \times p$ respectively. The prior mean vector $\boldsymbol{\mu}_0$ is of length n. The probability densities of $\boldsymbol{\theta}$ and $\mathbf{Y}$ are therefore:

$$\frac{(2\pi)^{-n/2}}{|\det \mathbf{V}_0|^{1/2}} \exp\left[-\frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\mu}_0)'\mathbf{V}_0^{-1}(\boldsymbol{\theta} - \boldsymbol{\mu}_0)\right]$$

and (6)

$$\frac{(2\pi)^{-p/2}}{|\det \boldsymbol{\Sigma}|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{Y} - \mathbf{H}\boldsymbol{\theta})'\boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \mathbf{H}\boldsymbol{\theta})\right]$$

in which $(\boldsymbol{\theta} - \boldsymbol{\mu}_0)'$ denotes the transpose of $(\boldsymbol{\theta} - \boldsymbol{\mu}_0)$, and the prior mean vector and dispersion matrix of $\boldsymbol{\theta}$ are $\boldsymbol{\mu}_0$ and $\mathbf{V}_0$, of sizes n and $n \times n$ respectively.

By Bayes' theorem, therefore, the posterior density of $\boldsymbol{\theta}$, given observations $\mathbf{Y} = \mathbf{g}$ will be proportional to

$$\begin{aligned}
&\exp\left[-\frac{1}{2}(\mathbf{g} - \mathbf{H}\boldsymbol{\theta})'\boldsymbol{\Sigma}^{-1}(\mathbf{g} - \mathbf{H}\boldsymbol{\theta})\right]\exp\left[-\frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\mu}_0)'\mathbf{V}_0^{-1}(\boldsymbol{\theta} - \boldsymbol{\mu}_0)\right] \\
&= \exp\left[-\frac{1}{2}(\mathbf{g}'\boldsymbol{\Sigma}^{-1}\mathbf{g} - \boldsymbol{\theta}'\mathbf{H}'\boldsymbol{\Sigma}^{-1}\mathbf{g} - \mathbf{g}'\boldsymbol{\Sigma}^{-1}\mathbf{H}\boldsymbol{\theta} + \boldsymbol{\theta}'\mathbf{H}'\boldsymbol{\Sigma}^{-1}\mathbf{H}\boldsymbol{\theta}\right. \\
&\qquad \left. + \boldsymbol{\theta}'\mathbf{V}_0^{-1}\boldsymbol{\theta} - \boldsymbol{\mu}_0'\mathbf{V}_0^{-1}\boldsymbol{\theta} - \boldsymbol{\theta}'\mathbf{V}_0^{-1}\boldsymbol{\mu}_0 + \boldsymbol{\mu}_0'\mathbf{V}_0^{-1}\boldsymbol{\mu}_0)\right].
\end{aligned} \tag{7}$$

This is of quadratic form and so may be re-written as

$$\exp\left[-\frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\mu}_1)'\mathbf{V}_1^{-1}(\boldsymbol{\theta} - \boldsymbol{\mu}_1)\right]$$

or

$$\exp\left[-\frac{1}{2}(\boldsymbol{\theta}'\mathbf{V}_1^{-1}\boldsymbol{\theta} - \boldsymbol{\mu}_1'\mathbf{V}_1^{-1}\boldsymbol{\theta} - \boldsymbol{\theta}'\mathbf{V}_1^{-1}\boldsymbol{\mu}_1 + \boldsymbol{\mu}_1'\mathbf{V}_1^{-1}\boldsymbol{\mu}_1)\right]. \tag{8}$$

Equating corresponding terms in (7) and (8) it may be seen that

$$\mathbf{V}_1^{-1} = \mathbf{V}_0^{-1} + \mathbf{H}'\boldsymbol{\Sigma}^{-1}\mathbf{H} \tag{9}$$

and

$$\mathbf{V}_1^{-1}\boldsymbol{\mu}_1 = \mathbf{V}_0^{-1}\boldsymbol{\mu}_0 + \mathbf{H}'\boldsymbol{\Sigma}^{-1}\mathbf{g}. \tag{10}$$

The posterior distribution of $\boldsymbol{\theta}$ is therefore MVN $(\boldsymbol{\mu}_1, \mathbf{V}_1)$. This completes the proof of the theorem.

The relations in (9) and (10) can be regarded then as updating the prior mean vector and dispersion matrix, $\boldsymbol{\mu}_0$ and $\mathbf{V}_0$ into the posterior mean vector and dispersion matrix $\boldsymbol{\mu}_1$ and $\mathbf{V}_1$. In fact the relationships above can be made simpler and more explicit by the

following: from (9)

$$\begin{aligned}\mathbf{V}_1 &= (\mathbf{V}_0^{-1} + \mathbf{H}'\boldsymbol{\Sigma}^{-1}\mathbf{H})^{-1} \\ &= \mathbf{V}_0 - \mathbf{V}_0\mathbf{H}'(\boldsymbol{\Sigma} + \mathbf{H}\mathbf{V}_0\mathbf{H}')^{-1}\mathbf{H}\mathbf{V}_0 \end{aligned} \tag{11}$$

(see, for example, Mardia *et al.*, 1979). Now, premultiplying (10) by $\mathbf{V}_1$ from (11):

$$\boldsymbol{\mu}_1 = (\mathbf{V}_0 - \mathbf{V}_0\mathbf{H}'(\boldsymbol{\Sigma} + \mathbf{H}\mathbf{V}_0\mathbf{H}')^{-1}\mathbf{H}\mathbf{V}_0)(\mathbf{V}_0^{-1}\boldsymbol{\mu}_0 + \mathbf{H}'\boldsymbol{\Sigma}^{-1}\mathbf{g})$$

which, after expanding and cancelling some terms, can be written as:

$$\boldsymbol{\mu}_1 = \boldsymbol{\mu}_0 + \mathbf{V}_0\mathbf{H}'(\boldsymbol{\Sigma} + \mathbf{H}\mathbf{V}_0\mathbf{H}')^{-1}(\mathbf{g} - \mathbf{H}\boldsymbol{\mu}_0). \tag{12}$$

So the prior distribution is MVN $(\boldsymbol{\mu}_0, \mathbf{V}_0)$ and the posterior distribution is MVN $(\boldsymbol{\mu}_1, \mathbf{V}_1)$, with the updating equations for the mean vector and dispersion matrix given by:

$$\boldsymbol{\mu}_1 = \boldsymbol{\mu}_0 + \mathbf{V}_0\mathbf{H}'(\boldsymbol{\Sigma} + \mathbf{H}\mathbf{V}_0\mathbf{H}')^{-1}(\mathbf{g} - \mathbf{H}\boldsymbol{\mu}_0) \tag{13}$$

and

$$\mathbf{V}_1 = \mathbf{V}_0 - \mathbf{V}_0\mathbf{H}'(\boldsymbol{\Sigma} + \mathbf{H}\mathbf{V}_0\mathbf{H}')^{-1}\mathbf{H}\mathbf{V}_0. \tag{14}$$

Note that if both dispersion matrices, $\boldsymbol{\Sigma}$ and $\mathbf{V}_0$, are scaled uniformly by the same factor, $\boldsymbol{\mu}_1$ is unaffected. The central calculation in (13) and (14) is the inversion of the symmetric matrix $\boldsymbol{\Sigma} + \mathbf{H}\mathbf{V}_0\mathbf{H}'$ which is of size $p \times p$. A variety of methods are available to achieve this inversion; see, for example Ayres (1962).

Even with such a method, however, it is worth investigating some special cases to see how the updating equations above may be simplified.

(1) *Independent observations*

Consider first the case in which there is just one observation $Y = g$; $p = 1$ and the matrix $\boldsymbol{\Sigma}$ consists of a single element τ^2. The matrix $\mathbf{H}$ becomes a row vector of length n, $\mathbf{h}'$ say. Therefore $\boldsymbol{\Sigma} + \mathbf{H}\mathbf{V}_0\mathbf{H}'$ is a scalar $\tau^2 + \mathbf{h}'\mathbf{V}_0\mathbf{h}$ and the inverse is the reciprocal. The updating equations then become:

$$\boldsymbol{\mu}_1 = \boldsymbol{\mu}_0 + \frac{\mathbf{V}_0\mathbf{h}}{\tau^2 + \mathbf{h}'\mathbf{V}_0\mathbf{h}}(g - \mathbf{h}'\boldsymbol{\mu}_0) \tag{15}$$

and

$$\mathbf{V}_1 = \mathbf{V}_0 - \frac{\mathbf{V}_0\mathbf{h}\mathbf{h}'\mathbf{V}_0}{\tau^2 + \mathbf{h}'\mathbf{V}_0\mathbf{h}}. \tag{16}$$

Now if the elements of the vector $\mathbf{h}'$ are $h_1, \ldots h_n$, and the elements of the prior mean vector $\boldsymbol{\mu}_0$ are $\mu_1^{(0)}, \ldots \mu_n^{(0)}$, the expected value of the observation is $\mathbf{h}'\boldsymbol{\mu}_0$ which is $\sum_{j=1}^{n} h_j\mu_j^{(0)}$, which we shall denote by v.

Furthermore letting the elements in the prior dispersion matrix $\mathbf{V}_0$ be $v_{ij}^{(0)}$, we shall denote $\sum_{j=1}^{n} h_j v_{ij}^{(0)}$ by S_i $(i = 1, \ldots n)$ and $\sum_{i=1}^{n} h_i S_i$ by T. Then it follows that the elements $\mu_i^{(1)}$, $v_{ij}^{(1)}$ $(i = 1, \ldots n; j = 1, \ldots n)$ in the posterior mean vector $\boldsymbol{\mu}_1$ and dispersion matrix $\mathbf{V}_1$ are given by:

$$\mu_i^{(1)} = \mu_i^{(0)} + \frac{S_i}{\tau^2 + T}(g - v) \quad (i = 1, \ldots n) \tag{17}$$

and

$$v_{ij}^{(1)} = v_{ij}^{(0)} - \frac{S_i S_j}{\tau^2 + T} \quad (i = 1, \ldots n; \quad j = 1, \ldots n). \tag{18}$$

The calculations in this special case of a single observation are therefore very simple. Now if the p observations are independent, the matrix $\mathbf{\Sigma}$ will consist of non-zero elements on the main diagonal only. Because of the sequential nature of Bayes' Theorem for independent observations, the method above in (17) and (18) can be applied sequentially to the individual observations (in any order); the posterior from one becomes the prior for the next, and so on.

(2) *Observations without error*

In some cases (especially in the turning flows problem) the observations may be made over such a long time that the resulting average flow estimates are virtually free of random error; that is $\mathbf{\Sigma} = 0$ and in (17) and (18) $\tau^2 = 0$. (Naturally, systematic error due to daily or seasonal variation may remain.) The observations may again be considered as independent, therefore, and the sequential method used, so that (17) and (18) become

$$\mu_i^{(1)} = \mu_i^{(0)} + \frac{S_i}{T}(g - v) \quad (i = 1, \ldots n) \tag{19}$$

and

$$v_{ij}^{(1)} = v_{ij}^{(0)} - \frac{S_i S_j}{T} \quad (i = 1, \ldots n; \quad j = 1, \ldots n). \tag{20}$$

Note again that $\boldsymbol{\mu}_1$ is invariant under uniform scaling of $\mathbf{V}_0$.

(3) *Least-informative prior*

Let the prior covariance matrix $\mathbf{V}_0$ be written as $\beta \mathbf{U}_0$ where β is a scalar and $\mathbf{U}_0$ is scaled so that the trace of the $p \times p$ matrix $\mathbf{H}'\mathbf{U}_0\mathbf{H}$ is p. Also let the error covariance matrix $\mathbf{\Sigma}$ be written as $\alpha\mathbf{\Gamma}$ where α is a scalar and $\mathbf{\Gamma}$ is scaled in the same way: trace $(\mathbf{\Gamma}) = p$. From (13)

$$\boldsymbol{\mu}_1 = \boldsymbol{\mu}_0 + \beta \mathbf{U}_0\mathbf{H}'(\alpha\mathbf{\Gamma} + \beta\mathbf{H}\mathbf{U}_0\mathbf{H}')^{-1}(\mathbf{g} - \mathbf{H}\boldsymbol{\mu}_0). \tag{21}$$

If β is allowed to tend to infinity, whilst α remains fixed, the prior distribution of $\boldsymbol{\theta}$ becomes "least-informative"; i.e. minimal confidence is placed on the prior. This corresponds to the idea of minimal information used by Van Zuylen and others. In these circumstances $\boldsymbol{\mu}_1 \to \boldsymbol{\mu}_1^*$ and $\mathbf{U}_1 \to \mathbf{U}_1^*$ (with $\mathbf{V}_1 = \beta\mathbf{U}_1$), where:

$$\boldsymbol{\mu}_1^* = \boldsymbol{\mu}_0 + \mathbf{U}_0\mathbf{H}'(\mathbf{H}\mathbf{U}_0\mathbf{H}')^{-1}(\mathbf{g} - \mathbf{H}\boldsymbol{\mu}_0) \tag{22}$$

and

$$\mathbf{U}_1^* = \mathbf{U}_0 - \mathbf{U}_0\mathbf{H}'(\mathbf{H}\mathbf{U}_0\mathbf{H}')^{-1}\mathbf{H}\mathbf{U}_0. \tag{23}$$

However, this result is exactly the same as that which would be obtained if $\alpha = 0$ regardless of the nature of $\mathbf{\Gamma}$ (and $\mathbf{\Sigma}$). Therefore $\boldsymbol{\mu}_1^*$ and $\mathbf{U}_1^*$ may be calculated by the sequential method since the observations $g_1, \ldots g_p$ can be introduced individually and independently by thinking of the covariance terms in $\mathbf{\Gamma}$ as all being zero. The minimal information or least-informative prior case, then, is particularly easy to apply, as it requires no matrix inversion even if the observations are not independent.

How close the posterior means $\boldsymbol{\mu}_1$ are to the "minimal information" means $\boldsymbol{\mu}_1^*$ depends on the ratio of the two scaling parameters α and β. If $\eta = \beta/(\alpha + \beta)$, then as $\eta \to 1$ $\boldsymbol{\mu}_1 \to \boldsymbol{\mu}_1^*$

whilst as $\eta \to 0$ $\boldsymbol{\mu}_1 \to \boldsymbol{\mu}_0$ (the posterior becomes identical to the prior). Note also that as $\eta \to 1$, $\mathbf{H}\boldsymbol{\mu}_1 \to \mathbf{H}\boldsymbol{\mu}_1{}^* = \mathbf{g}$ (the observations act as linear equality constraints).

Before demonstrating some aspects of the application of the method by means of some examples, some general points of practical interest should be mentioned. First it is envisaged that the prior dispersion matrix $\mathbf{V}_0$ will be diagonal; i.e. the θ_i will be independent (except, of course, when $\mathbf{V}_0$ is taken from the posterior distribution in the previous stage in the sequential method). Second, if necessary the prior means will have been scaled up so as to be comparable with the magnitude of the observations; i.e. the observed total flow is equal to the prior estimate of this total flow

$$\sum_{i=1}^{p} g_i = \sum_{i=1}^{p} \sum_{j=1}^{n} h_{ij}\mu_j. \tag{24}$$

The successful operation of the method does not depend upon this but it seeems sensible to do this initially, especially in cases where the prior information consists largely of beliefs about the *relative* magnitude of the parameters and where there could well have been some overall growth in the magnitude of the parameters (as with an out-of-date transportation survey) since the prior information was collected. Third, since the prior information will often have been taken from counts (as with short manual counts at the intersection in the turning flows problem), the variances in the matrix $\mathbf{V}_0$ may be estimated by assuming that the counts are Poisson random variables, for which the variance equals the mean. Thus if a count x_i were scaled into, say, an hourly flow rate of $\mu_i(=\gamma x_i)$, taking account also of any uniform scaling on the priors described above, then the variance of the prior will be estimated as $\gamma^2 x_i$.

EXAMPLES

(1) *Turning flows at an intersection*

Suppose that at a four-arm intersection, the inflows and outflows on the approach roads are, respectively (520, 1093, 625, 447) and (731, 405, 540, 1009). The prior means of the turning flows are

		To			
		1	2	3	4
From	1	0	25	381	97
	2	61	0	61	904
	3	611	72	0	18
	4	116	301	38	0

The prior variances will be assumed to be proportional to the means, with zero covariances. (It should be noted that the prior means are consistent with the observations in that the sum of the prior means is 2685, the same as the total inflow and the total outflow.) Assuming that the inflow and outflow rates are based on long term counts, the matrix $\boldsymbol{\Sigma}$ will be taken to be zero, and the sequential method of case 2 will be applied. Table 1 below shows the posterior means after the introduction of each new observation.

The data above were based on the example considered by Norman, Hoffman and Harding (1979), who gave results produced by the entropy maximisation method. These results are also shown at the foot of Table 1. It can be seen that there is close agreement between the two methods. This is not surprising since, as was mentioned earlier, when the prior variances are large compared with the observation variances, the results reflect the minimal use of the prior information as in the information minimising and entropy maximising approaches.

Note that in such applications, if a particular turning flow is banned or impossible then $\mu_i = \mathrm{Var}(\theta_i) = 0$ for that flow. Similarly if an accurate long term count is available for some turning flow then again $\mathrm{Var}(\theta_i)$ should be set to zero for that flow in the matrix $\mathbf{V}_0$.

Table 1. Estimated mean turning flow after introduction of each observation

Iteration	T_{12}	T_{13}	T_{14}	T_{21}	T_{23}	T_{24}	T_{31}	T_{32}	T_{34}	T_{41}	T_{42}	T_{43}	Observation
0	25	381	97	61	61	904	611	72	18	116	301	38	
1	26	394	100	61	61	904	611	72	18	116	301	38	$a_1 = 520$
2	26	394	100	65	65	963	611	72	18	116	301	38	$a_2 = 1093$
3	26	394	100	65	65	963	545	64	16	116	301	38	$a_3 = 625$
4	26	394	100	65	65	963	545	64	16	114	296	37	$a_4 = 447$
5	26	394	100	67	65	961	547	62	16	117	293	37	$b_1 = 731$
6	31	390	99	75	64	953	544	68	13	112	306	30	$b_2 = 405$
7	29	419	72	82	85	927	543	71	10	106	305	36	$b_3 = 540$
EM method	29	416	75	84	88	921	542	70	13	105	306	36	

(2) *Trip matrix estimation*

The small illustrative example described here has been taken from Bell (1983). There are 6 O–D pairs and 4 links on which observations on the flow are made. Suppose that a proportional assignment algorithm gives

$$\mathbf{H} = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0.3 & 1 & 1 & 0 & 0 & 0 \\ 0.3 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

and observations on the link flows on several occasions produce mean observations

$$\mathbf{g} = (19.2, 20.8, 10.0, 13.0)'$$

and a dispersion matrix

$$\boldsymbol{\Sigma} = \begin{bmatrix} 7.04 & 1.41 & 1.65 & 0.65 \\ 1.41 & 2.84 & 1.25 & -0.60 \\ 1.65 & 1.25 & 1.60 & 0.10 \\ 0.65 & -0.60 & 0.10 & 0.50 \end{bmatrix}.$$

Since the trace of $\boldsymbol{\Sigma}$ is 11.98, $\alpha = 2.995$. The prior means μ_i are all taken initially to be 10.0, with variances also of 10.0, but since

$$\sum_{i=1}^{p} \sum_{j=1}^{n} h_{ij}\mu_j \quad \text{is } 106.0$$

whilst $\sum_{i=1}^{p} g_i$ is only 63.0, the μ's are all scaled by a factor of 0.594. Therefore,

$$\boldsymbol{\mu}_0 = (5.94 \quad 5.94 \quad 5.94 \quad 5.94 \quad 5.94 \quad 5.94)'$$

and $\mathbf{V}_0$ is of diagonal form with all six elements on the main diagonal equal to 3.53.

The trace of $\mathbf{HV}_0\mathbf{H}'$ is 35.96 so that $\beta = 8.99$. Therefore $\eta = 0.750$.

From (13) and (14) the posterior means and variances of the $\boldsymbol{\theta}$ are found to be

$$(9.88 \quad 3.43 \quad 4.75 \quad 5.09 \quad 6.41 \quad 7.27)'$$

and

$$(1.64 \quad 1.76 \quad 1.88 \quad 1.85 \quad 1.76 \quad 2.45)' \text{ respectively.}$$

By allowing β to tend to infinity, the prior becomes least-informative and $\boldsymbol{\mu}_1{}^*$ is found from (22) to be

$$(15.43 \quad 1.09 \quad 4.28 \quad 2.59 \quad 5.78 \quad 9.14)'$$

which shows quite close agreement with the entropy maximising solution obtained by Bell to the same problem:

$$(15.43 \quad 2.06 \quad 3.32 \quad 3.20 \quad 5.17 \quad 10.72)'.$$

Figure 1 shows how the posterior means change as η moves from 0 (posterior equals prior) to 1 (least-informative prior). It can be seen that the $\boldsymbol{\mu}_1$ are most sensitive to variations in η in the neighbourhood of $\eta = 1$.

Naturally with a different prior a different posterior emerges, as shown in Fig. 2. In this the unscaled prior means were

$$(20.0 \quad 4.0 \quad 6.0 \quad 8.0 \quad 10.0 \quad 10.0)'$$

with, again, the variance to mean ratios taken to be unity. It should be noted that when

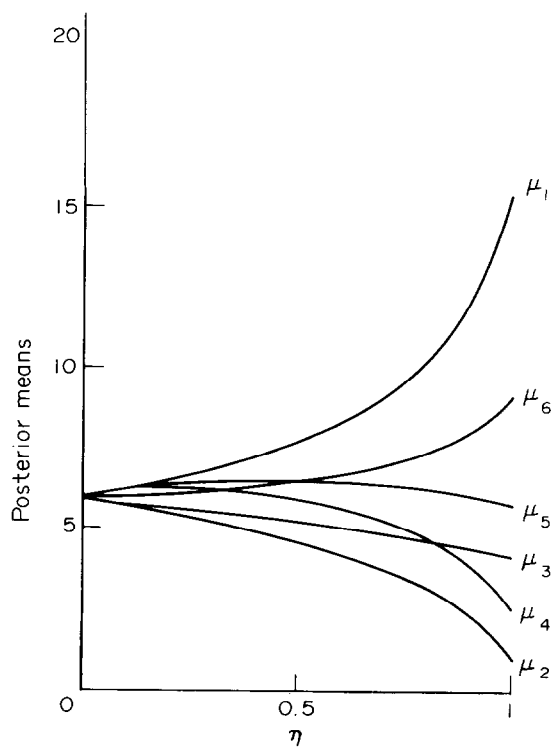

Fig. 1. Variation in the posterior means for the six O–D pairs, from the prior mean values ($\eta = 0$) to the least-informative prior (or minimal information) case ($\eta = 1$). The prior means are all equal.

$\eta = 1$, $\boldsymbol{\mu}_1{}^*$ is:

$$(15.43 \quad 1.47 \quad 3.90 \quad 2.46 \quad 5.91 \quad 9.39)'$$

slightly different from the $\boldsymbol{\mu}_1{}^*$ shown in Fig. 1. For, of course, although when $\eta = 1$ the *minimal* use is being made of the prior, this is not to say that no use is being made of the information there. As may be seen from (22), $\boldsymbol{\mu}_1{}^*$ is still a function of $\boldsymbol{\mu}_0$. Even if it is thought proper, then, to place as little degree of belief in the prior as possible, it is nevertheless important to use as good and accurate a prior as one can.

The posterior means for the least-informative prior case are also dependent upon the prior variances, as may be seen from the following. Maintaining the uniform prior means of 10.0 each (before scaling) but changing the variance to mean ratios from unity to (1.0 0.7 1.3 0.7 1.3 1.3) produces almost exactly the same posterior means as those given by Bell's maximum entropy method. One interpretation, amongst others, of this is that the maximum entropy method, although not *explicitly* stating anything about the relative degrees of belief in the different parts of the prior, does *implicitly* place values on them.

SENSITIVITY ANALYSIS

The dependence of $\boldsymbol{\mu}_1$ on the value of η shown in Fig. 1 was obtained by running a computer program to output $\boldsymbol{\mu}_1$ many times with different values of η. In practice this would be a time-consuming process and yet it is very desirable to have some idea of how the solution obtained would be affected by a change in one or more of the input parameters. When these changes are small, some relatively simple results can be obtained showing the sensitivity of the solution to changes in, for example, η (and hence α and β), $\boldsymbol{\mu}_0$, $\mathbf{H}$ and $\mathbf{g}$.

By differentiating (21), it can be shown that

$$\frac{\mathrm{d}\boldsymbol{\mu}_1}{\mathrm{d}\eta} = \mathbf{U}_0\mathbf{H}'[(1-\eta)\boldsymbol{\Gamma} + \eta\mathbf{H}\mathbf{U}_0\mathbf{H}']^{-1}\boldsymbol{\Gamma}[(1-\eta)\boldsymbol{\Gamma} + \eta\mathbf{H}\mathbf{U}_0\mathbf{H}']^{-1}(\mathbf{g} - \mathbf{H}\boldsymbol{\mu}_0) \tag{25}$$

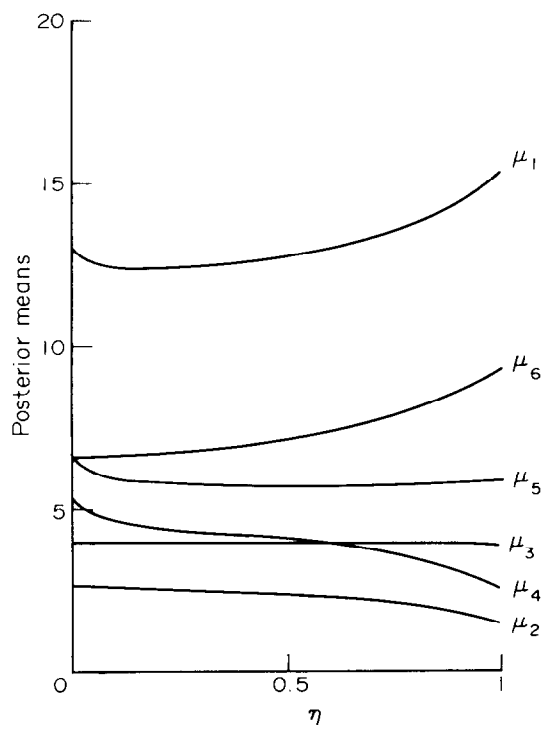

Fig. 2. As in Fig. 1, but with unequal prior means.

and, therefore, at the same time as the posterior means are calculated, by means of the inversion of the matrix $[(1-\eta)\mathbf{\Gamma} + \eta \mathbf{H}\mathbf{U}_0\mathbf{H}']$, this gradient vector can be found also. Hence, if the assumed values of α or β were to be altered by some small amount, the small change in η, $\Delta\eta$, can be calculated, and the small change in $\boldsymbol{\mu}_1$ is given, approximately, by

$$\Delta\boldsymbol{\mu}_1 = \Delta\eta \frac{\mathrm{d}\boldsymbol{\mu}_1}{\mathrm{d}\eta}. \tag{26}$$

The effect of a small change in the prior means $\boldsymbol{\mu}_0$ can be calculated by noting that the posterior means $\boldsymbol{\mu}_1$ are a linear combination of the prior means. Ignoring the second order effect of the rescaling of the $\boldsymbol{\mu}_0$ which would result (if scaling had been done), (13) shows that

$$\Delta\boldsymbol{\mu}_1 = [\mathbf{I} - \mathbf{V}_0\mathbf{H}'(\mathbf{\Sigma} + \mathbf{H}\mathbf{V}_0\mathbf{H}')^{-1}\mathbf{H})\Delta\boldsymbol{\mu}_0. \tag{27}$$

It has been assumed that the matrix $\mathbf{H}$, which contains the results of the proportional assignment process, is known exactly. In practice there may be some doubt about the precise value of some elements in this matrix, and so it would be wise to investigate the sensitivity of the posterior means to small changes in an element h_{rs}, say. Now h_{rs} is the proportion of trips between O–D pair s which use link r. Consider therefore the contribution to $\boldsymbol{\mu}_1$ made by the observation on link r. To do this, let us make the approximation that the observations are independent so that the sequential method could be applied, and assume that observation r was the last to be introduced. Then by (17)

$$\mu_i^{(1)} = \mu_i^{(0)} + \frac{S_i(g_r - v)}{\tau^2 + T} \quad (i = 1, \ldots n) \tag{28}$$

where $S_i = \sum_j h_{rj}v_{ij}^{(0)}$; $T = \sum_i \sum_j h_{ri}h_{rj}v_{ij}^{(0)}$; $v = \sum_j h_{rj}\mu_j^{(0)}$ and τ^2 is the variance of this observation, whose value is g_r. Now in this the superscript 0 refers, strictly, to the prior immediately before this observation r is introduced, and the superscript 1 to the final posterior. However, if there are many observations, the influence of a single observation will not be too great and so, for example, $v_{ij}^{(0)}$ can be replaced by $v_{ij}^{(1)}$ without too much error. Then by differentiating with respect to h_{rs}:

$$\frac{\mathrm{d}\mu_i}{\mathrm{d}h_{rs}} = \frac{S_i(g_r - v)}{\tau^2 + T}\left[\frac{v_{is}}{S_i} - \frac{2S_s}{\tau^2 + T} - \frac{\mu_s}{g_r - v}\right] \quad (i = 1, \ldots n). \tag{29}$$

This allows us to estimate the size of the changes in $\boldsymbol{\mu}_1$, albeit rather crudely, caused by a change in a single element of $\mathbf{H}$. It is likely that modifications to the proportional assignment process will result in more than one element of $\mathbf{H}$ being altered. Their effects on each μ_i are however additive.

In the same way and with the same notation the effect of a change in g_r on $\boldsymbol{\mu}_1$ can be shown to be determined by:

$$\frac{\mathrm{d}\mu_i^{(1)}}{\mathrm{d}g_r} = \frac{S_i}{\tau^2 + T} \quad (i = 1, \ldots n). \tag{30}$$

The approximate results above, then, give some idea of the sensitivity of the solution to small changes in the various parts of the input parameters and data. All of these can be calculated at the end of the process of determining the posterior means, rather than having to re-run the complete program again with different parameter values.

The inclusion of extra observations on link flows always improves the precision of the estimation of the trip matrix in that the posterior dispersion matrix $\mathbf{V}_1$ is reduced. Since not all link flows in a network are ever likely to be measured, it is worth investigating whether it is possible to set out rules or guidelines for determining which link flows should be measured. In keeping with the levels of approximation used in the sensitivity analyses

above, assume that from the link flow observations used so far the posterior means and dispersion matrix are $\boldsymbol{\mu}_0$ and $\mathbf{V}_0$. A link on which it is proposed to make a flow measurement has coefficients $h_1, \ldots h_n$ representing, as usual, the proportions of flow between the n O–D pairs, which use that link. Then, from (17) the reduction in the element v_{ij} in the dispersion matrix will be $S_i S_j/(\tau^2 + T)$, where $S_i = \sum_j h_j v_{ij}^{(0)}$, $T = \sum_i \sum_j h_i h_j v_{ij}^{(0)}$ and τ^2 is the (estimated) error variance of this observation, taking account of the proposed length of time over which this observation is to be made. Then a sensible overall measure of the benefit arising from this observation will be the total reduction in the variances of the θ_i: i.e., the reduction in the trace of the dispersion matrix, $\sum_i S_i^2(\tau^2 + T)$. The guideline, then, for choosing which links on which to make further observations is to select those links which give the highest values of the measure above.

SUMMARY AND CONCLUSIONS

The problem of estimating a trip matrix in which there are n elements $\theta_1, \ldots \theta_n$ by means of observations on p link flows ($p < n$) is underspecified; prior beliefs about the $\boldsymbol{\theta}$ must be added to allow a unique solution to be found. Previous work had used the closely related principles of minimal information and maximum entropy to enable the problem to be posed as an optimisation problem. Neither of these approaches allows anything other than minimal belief in the prior; the point prior estimates on the $\boldsymbol{\theta}$ are used simply in the form of a "target" trip matrix, typically, so that the trip matrix is found which is "closest" to this target, subject to certain constraints. The observations are taken to be without error so that they are used to given the constraints.

From the standpoint of Bayesian inference, the approach above is just one extreme case from a continuum of possibilities. Prior beliefs are modified by observations to produce posterior beliefs; the stronger the prior beliefs the less influence the observations will have in determining the posterior beliefs. The posterior beliefs, then, are a weighted average of the prior beliefs and the observations, and the relative weights in this average are determined by the relative magnitudes of the two dispersion matrices: $\boldsymbol{\Sigma}$ for the observation errors and $\mathbf{V}_0$ for the prior beliefs. The method proposed in this paper has the advantage, then, of allowing much more flexibility than previous methods in the degree of prior beliefs. Furthermore it allows for different degrees of belief in different parts of the prior. The "least-informative prior" or "minimal information" case is seen, therefore, to be just one extreme case of a whole range of possibilities.

From the computational viewpoint, the method is attractive because the assumptions of proportional assignment and multivariate normality are highly compatible, giving rise to a simple updating scheme for the means $\boldsymbol{\mu}$ and dispersion matrix $\mathbf{V}$. For independent observations the posterior from one stage becomes the prior for the next, enabling the easy introduction of fresh observations to modify the present estimates. The ease with which the analysis can be carried out depends on two assumptions: multivariate normality of the counts and accurate knowledge of the matrix $\mathbf{H}$ containing the results of the assignment process. The normality assumption is probably the more serious of these as many cells in the O–D matrix may have small mean numbers of trips and therefore it is in these cells that the normal approximation may break down. An alternative approach might be to assume that the counts were independent Poisson random variables, whilst the prior distributions of the $\boldsymbol{\theta}$ were gamma, as in Maher (1983). However the posterior distribution of the $\boldsymbol{\theta}$ is then a multivariate form of gamma, and not at all easy to use. The necessity of assuming accurate knowledge of $\mathbf{H}$ can be counteracted by the incorporation of sensitivity analysis into the computational process, so that the effect of small changes in the assumed values of the parameters on the solution can be investigated without rerunning the entire program.

A computer program has been written to implement the method and results from this show, as expected, that there is broad agreement between the "least informative prior" case and the maximum entropy and minimal information methods of, for example, Willumsen

and Van Zuylen. The closeness of this agreement is influenced by both the prior means and the prior variances.

Although no detailed tests have yet been carried out, it is thought that the computational effort required in the two approaches is roughly comparable, with the Bayesian method using more storage but less computation time than the maximum entropy or minimum information approach.

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